

Class-X ch-3 [Geography] * Water Resources

• Introduction

→ $\frac{3}{4}$ of Earth Surface is covered with water [But only a small part is usable]??

→ usable part : * Surface Run off * Groundwater



• Renewed by Hydrological Cycle.

Q. Then why water is scarce?? 😞??

→ 96.5% of total volume of water is in oceans and only 2.5% is fresh water.

→ 70% fresh water occurs as ice sheet and glaciers.

→ India receives nearly 4% of global precipitation and ranks 133 in terms of water availability per person per annum.

→ Total renewable water Resources of India are estimated at 1897 km^3 per annum.

→ By 2025, large part will face absolute water scarcity.

Class-X ch-3 [Geography] * Water Resources

• Water Scarcity - I

→ Abundance and Renewability of water ⇒ How Can it be Scarce??

We think →
↓
[It May be in areas of desert and drought
affected regions. [x]

• Caused Mainly due to

→ over exploitation, Excessive use
and unequal access to water among
different group.

Falken Marks
Water stress occurs when availability
of water is between 1,000 and 1,600 Cubic
Meter per person per year

Class-x ch-3 [Geography] • Water Resources

• Water Scarcity = II

→ quantitative aspect

Cities

Population ↑

Demand for water ↑ ⊕ food ↑

More irrigation ↑

- Exploitation of water Resources and falling ground water levels.

!! having their own ground !!
water pumping device

Urbanisation and Industrialisation

Heavy use of water
⊕

Power Consumer
[Hydro electricity]

* 22% of electricity
!! required is produced by !!
Hydroelectric plant.

→ qualitative aspect

↓
quantity is Sufficient but
area still suffer from water
Scarcity ??

↓
Due to Bad quality of water

↓
Polluted due to Domestic and
Industrial wastes, chemicals
Pesticide and fertilisers used. !!
Hazardous for Human use

Class-x ch-3 [Geography] • Water Resources

Need for Water Conservation and Management

→ To safeguard ourselves from ÷

↓
• Health hazards

[qualitative aspect]

• Ensure food security

[growing population]

• Degradation of natural ecosystem

• Water Scarcity will degrade natural resources and cause Ecological Crisis, affecting our lives negatively.

Class-X ch-3 [Geography] • Water Resources

• Multi Purpose River Project and Integrated Water Resources Management - I

→ How do we Conserve and Manage ??

• Historical and Archaeological records of water Conservation :

- (i). First Century B.C Srngaverapura had Sophisticated water harvesting System channelling flood water of river ganga.
- (ii). Dams, lakes and irrigation System were Extensively built during Chandragupta maurya.
- (iii). Evidences of Sophisticated irrigation works have also been found in Kalinga (Orissa), Nagarjunakonda (A.P) Beemur (Karnataka), Kolhapur (Maharashtra)

(iv). In the 11th Century, Bhopal lake one of the largest artificial lakes of its time was built

(v). In 14th Century, the tank in Hauz Khas, Delhi was Constructed by Iltutmish for supplying water to Siri fort area.

Class-X ch-3 [Geography] • Water Resources

• Multi-purpose River Project and Intergated Water Resoure Mangement.

→ Dam = It is a barrier across flowing water that obstructs, directs or retards the flow of water, often Creating a reservoir lake or impoundments.

• Based on structure ex: Timber dam • Based on Heights ex: Small scale

→ Why MPP ??

- Irrigation
- Electricity generation
- Flood Control
- Recreation
- Inland navigation
- Fish breeding

“Temples of Modern India”

- J. L Nehru

- Development of agriculture and village economy.
- Rapid Industrialisation and growth of urban economy.

= Integrated use of Impounded water

Class - X ch-3 [Geography] • Water Resources

• Opposition of Multipurpose projects - I

→ Geographical Reason

- Regulating or damming rivers affect natural flows.

- Excessive Sedimentation at the bottom.

Resulting rockier stream beds

- poorer habitats for aquatic life,

Submergence of existing vegetation and soil.

Induced earthquakes

Dams

• Unsuccessful in controlling Floods at the time of rainfall.

Ex: Release of water from dams

• Destruction of property, soil erosion and land degradation caused water borne diseases and pests

Class-X ch-3 [Geography] • Water Resources

• Opposition of Multipurpose projects - II

→ Social Impact

- large Scale Displacement unfair share of Sacrifice
- Irrigation → Changed Cropping pattern

• Salinisation of Soil

• Transformed Social landscape
→ [Gap among Rich and Poor ↑]

- Social Movement against MPP's
"Narmada Bachao Andolan" and "Tehri Dam Andolan."

- Same water Resources
different use. ?? 😞??

→ Agitation by farmers in
Gujarat, Sabarmati basin.
over water supply priority
given to urban areas.

→ Inter State water
dispute.

Class-X ch-3 [Geography] • Water Resources

• Rain water Harvesting

- Economically and Environmentally viable alternative in period of resistance against MPP.
- Variation in water harvesting system, keeping the local ecological conditions and their water needs in mind.

(i). Gul's and Kuls

- In mountainous region of western Himalayas impoundment on hill tops.

(ii). Khadins and Johads

- Jaisalmer and Rajasthan
- In arid area agricultural fields were converted into rain fed storage structures

(iv). Inundation channels in flood plains of Bengal.

(v). Roof top Rain water Harvesting.

(iii). Bamboo drip irrigation System

- In Meghalaya, 200 year old system of using bamboo pipe to transport water.
- 18-20 liters of water enters the bamboo pipes and 20-80 drop reaches at the site of plants.

Class-X ch-3 [Geography] • Water Resources

• Roof top Rain water Harvesting

→ In arid and semi-arid area of Rajasthan [Bikaner, Phalodi]



• Had Tankas [underground]



• Connected with Rooftop Rainwater Harvesting.



• Through pipes water is transported to tankas.

② Reliable Source of Drinking water.

② Beat the Summer Heat.

Tanka System



Grondathur Model

→ In, Mysore Karnataka 200 Household have installed Rooftop Rain water Harvesting System.

→ 1,000 mm annual precipitation with 80% Collection efficiency, 50000 liters of water is collected by every household annually.

* Tamil Nadu is the first state having Compulsory legal provision for Rooftop rain water harvesting.

Class-x ch-3 [Geography] Water Resources

• Frequently asked Questions

- (i). Water Scarcity? Causes of it? Specific Cause? Need for Conservation and Management of water resources?
- (ii). Ancient water resource management System?
- (iii). What is Dam? Why it is known MPP? Advantages of it?
- (iv). Reasons for opposition of MPP's? Problems arising due to MPP's? [specific]
- (v). Various Methods of Rain water harvesting?
- (vi). Rooftop rainwater harvesting? Gendathun model?
- (vii). Define: Dam, Guls and kuls, Khadin and Johads, Bamboo drip irrigation etc.